

POZNAN LAWICA, Poland

WMO: 123300

Lat: 52.4167N

Lon: 16.8347E

Elev: 88

StdP: 100.27

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-10.3	-16.2	0.9	-12.7	-13.2	1.2	-9.1	12.3	4.9	10.8	4.0	2.7	80	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	30.8	19.5	28.8	18.8	27.0	18.1	20.9	27.9	20.1	26.4	19.2	24.8	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	22.7	18.0	13.1	22.0	17.1	12.3	21.5	60.7	27.8	57.8	26.6	54.9	24.8	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.5	7.5	DB	-16.2	34.0	4.3	1.8	-19.3	35.3	-21.8	36.3	-24.2	37.3	-27.3	38.6	(4)
(5)			WB	-16.5	22.7	4.2	1.0	-19.5	23.3	-22.0	23.9	-24.4	24.4	-27.5	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-0.8	0.5	3.8	9.4	13.9	17.2	19.5	19.0	14.4	9.3	4.4	0.8	(6)
(7)		DBStd	8.23	5.17	4.87	4.23	4.30	3.99	3.62	3.34	3.14	3.34	3.93	3.80	4.66	(7)
(8)		HDD10.0	1386	334	267	196	62	9	0	0	0	3	61	169	285	(8)
(9)		HDD18.3	3444	592	500	452	269	146	63	26	29	126	280	417	543	(9)
(10)		CDD10.0	1143	0	0	3	43	131	217	293	281	134	39	2	0	(10)
(11)		CDD18.3	160	0	0	0	1	9	30	61	52	7	0	0	0	(11)
(12)		CDH23.3	1591	0	0	0	21	118	318	599	466	69	1	0	0	(12)
(13)		CDH26.7	461	0	0	0	3	21	93	202	129	13	0	0	0	(13)
(14)	Wind	WSAvg	3.6	4.0	3.9	4.0	3.7	3.4	3.4	3.3	3.0	3.2	3.3	3.6	3.9	(14)
(15)	Precipitation	PrecAvg	546	36	30	39	31	53	61	84	58	43	35	35	39	(15)
(16)		PrecMax	721	75	64	82	79	107	115	187	147	110	94	99	99	(16)
(17)		PrecMin	362	4	6	9	7	11	6	14	10	7	6	1	7	(17)
(18)		PrecStd	85	19	14	19	18	25	31	46	29	24	23	18	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.3	17.7	26.1	29.0	32.7	33.2	32.6	28.5	22.0	14.6	11.8	(19)
(20)			MCWB	8.8	8.7	10.8	15.2	18.1	20.8	20.6	20.4	18.4	15.9	11.9	9.9	(20)
(21)		2%	DB	8.8	10.1	14.8	22.6	26.7	29.2	31.1	29.9	25.2	19.3	12.8	9.7	(21)
(22)			MCWB	7.3	7.7	9.2	13.7	16.8	18.9	19.7	19.4	17.0	14.6	10.5	8.2	(22)
(23)		5%	DB	7.1	8.6	12.2	19.7	24.1	26.9	28.9	27.8	23.0	17.2	11.1	8.2	(23)
(24)			MCWB	5.7	6.5	8.4	12.2	15.8	17.9	19.0	18.7	16.1	13.7	9.4	6.9	(24)
(25)		10%	DB	5.8	7.0	10.2	17.2	21.9	24.8	26.8	26.0	20.8	15.1	9.9	6.7	(25)
(26)			MCWB	4.6	5.3	7.1	11.0	14.6	16.8	18.6	18.0	15.2	12.6	8.5	5.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.8	12.0	16.0	19.6	22.0	22.0	21.7	19.4	16.7	12.6	10.3	(27)
(28)			MCDWB	10.3	11.3	16.9	24.1	26.4	29.2	29.3	28.5	25.9	20.6	14.1	11.5	(28)
(29)		2%	WB	7.3	8.1	10.2	14.4	17.9	20.1	21.0	20.7	18.1	15.2	10.9	8.4	(29)
(30)			MCDWB	8.6	9.7	13.4	21.3	24.0	27.1	27.9	27.2	23.3	18.3	12.2	9.5	(30)
(31)		5%	WB	5.8	6.6	8.8	12.9	16.7	18.7	20.2	19.8	16.9	14.0	9.7	7.0	(31)
(32)			MCDWB	6.9	8.2	11.8	18.3	22.2	24.5	26.6	25.5	21.4	16.6	10.9	8.1	(32)
(33)		10%	WB	4.5	5.3	7.4	11.6	15.5	17.7	19.3	18.9	15.8	12.9	8.5	5.7	(33)
(34)			MCDWB	5.5	6.9	9.9	16.1	20.6	22.8	25.0	23.9	19.7	15.0	9.7	6.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	5.9	7.8	10.2	10.6	10.5	10.2	10.6	9.4	7.7	5.2	4.4	(35)
(36)			MCDBR	5.5	7.5	11.5	13.9	14.0	13.8	13.9	13.8	13.1	10.4	7.1	5.3	(36)
(37)			MCWBR	4.6	5.6	7.2	6.9	6.2	5.8	4.9	5.4	6.1	6.3	5.4	4.5	(37)
(38)		5% WB	MCDWB	5.4	7.1	10.4	12.4	12.0	12.0	11.8	11.8	11.1	9.3	6.7	5.1	(38)
(39)			MCWBR	4.7	5.6	7.0	6.5	6.1	5.8	4.8	5.2	5.9	6.0	5.5	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.342	0.377	0.420	0.415	0.411	0.436	0.430	0.388	0.367	0.344	0.310	(40)	
(41)		taud	2.303	2.296	2.238	2.174	2.237	2.277	2.221	2.246	2.336	2.367	2.352	2.324	(41)	
(42)		Ebn at Noon	667	759	802	811	837	843	811	794	785	722	627	605	(42)	
(43)		Edh at Noon	63	85	110	132	130	126	132	122	99	79	61	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.67	1.35	2.50	4.16	5.11	5.35	4.99	4.40	3.18	1.75	0.77	0.48	(44)	
(45)		RadStd	0.10	0.23	0.33	0.54	0.58	0.47	0.61	0.40	0.40	0.24	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	-184	+107	+38
(47) Regional (28 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+66	N/A

Nomenclature: See separate page